MIL-B-51492A(CG)
20 November 1987
SUPERSEDING
MIL-B-51492(CG)
23 August 1982

MILITARY SPECIFICATION

BELT, SAFETY, BOAT CREW

This specification is approved for use within the U.S. Coast Guard, Department of the Navy, and is available for use by all Departments and and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for an adjustable nylon-web safety belt, with quick release type buckle, for use by personnel engaged in hazardous boat operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-T-295 - Thread, Nylon
QQ-P-416 - Plating, Cadmium (Electrodeposited)

MILITARY

MIL-W-4088 - Webbing, Textile, Woven Nylon

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commandant (G-OSR-2), U.S. Coast Guard, Washington, DC 20593 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, NY 10016.)

ASTM STANDARDS

B 117 - Salt Spray (Fog) Testing

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:02Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction sample. Prior to the time of the award, the contractor shall submit to the contracting officer for approval a finished preproduction sample belt conforming to the requirements of this specification.

3.2 Material and components. Material shall be free from all defects and imperfections that might affect the serviceability and appearance of the finished product. Materials and components shall conform to applicable specifications and standards as listed or required herein. Materials and components which are not covered by applicable specifications or standards, or are not specifically described herein shall be of the best quality, of the lightest practical weight, and entirely suitable for the purpose intended.

3.2.1 Holding straps (Item 6). Except for color, the holding straps shall be nylon webbing in conformance with type X of MIL-W-4088, with a minimum breaking strength of 8700 pounds. ("Item" as used here and in subsequent paragraphs refers to the component numbering of figure 1).

3.2.2 Body belt (Item 7). The body belt shall be nylon webbing in conformance with type XXIV of MIL-W-4088, with a minimum breaking strength of 5500 pounds.

3.2.3 Thread. The thread used to stitch the boat crew safety belt shall be nylon conforming to type II, class A, size 6 of V-T-295, and shall be of equal color to the webbing.

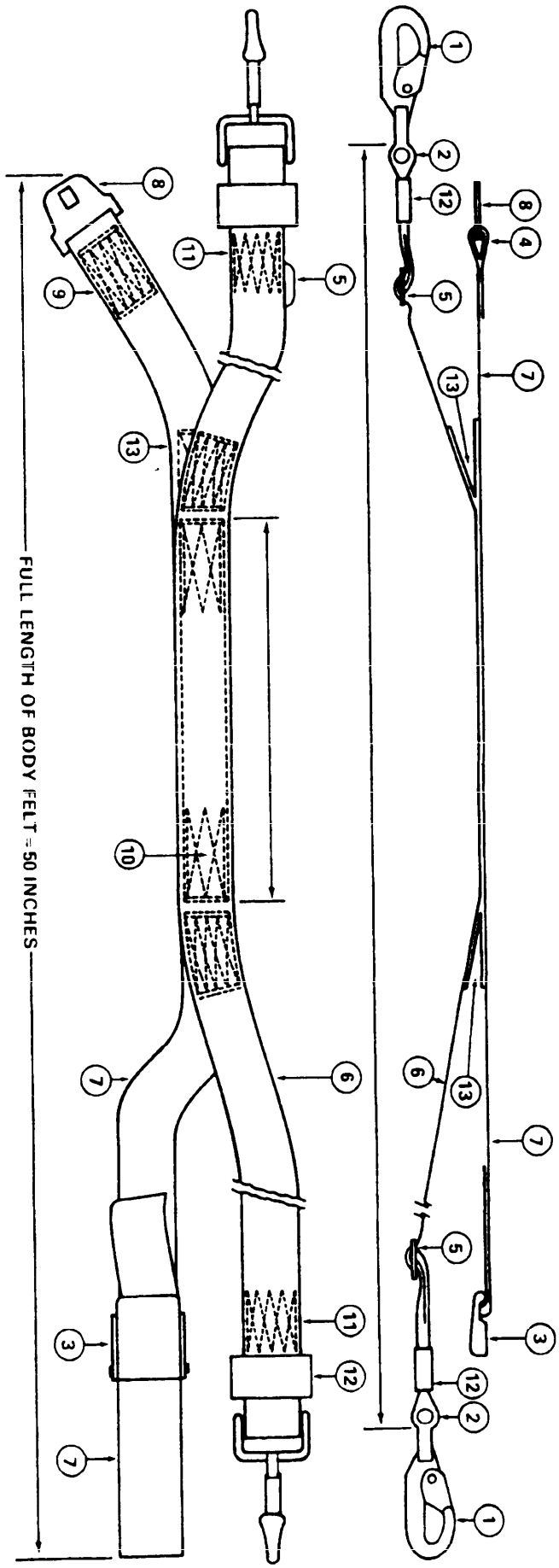
3.2.4 Metal Parts (Items 1, 2, 3, 5, and 8). All metals shall be of a corrosion resisting type or shall be cadmium plated in accordance with QQ-P-416. Snap hooks and all buckles shall be capable of withstanding a tensile load of 1500 pounds without cracking, breaking, or taking a permanent deformation. Snap hooks and all buckles shall be capable of withstanding a tensile load of 2000 pounds without cracking or breaking. All metal parts shall be free of burrs and sharp edges. Bidders shall indicate the type of metal to be used (see 6.3).

3.2.5 Reinforcements (Item 13). Snap reinforcements shall be similar in color and material to the holding strap.

3.3 Design and construction. The belt shall be designed and constructed in accordance with figure 1 and as further specified herein.

3.3.1 Dimensions. Dimensions of all component parts of the safety belt shall be in accordance with figure 1 and as further specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.



ITEM	DESCRIPTION
1	Snap Hook
2	Bushing
3	Body Belt Buckle Quick Release
4	Insert Reinforcing
5	Buckles, Slide
6	Strap, Holding 92" x 1 3/4"
7	Belt, 2" x 50"
8	Body Belt Buckle Male End
9	Stitching Hook & Buckle
10	Stitching Body Pad
11	Stitching Strap End
12	Keepers, Slide
13	Reinforcement Strip

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

3.3.2 Stitching. All stitching shall be lock stitch, type 301 of FED-STD-751. The pattern of stitching used to attach the male end of the buckle to the body belt shall be a four point cross stitch with a minimum length of 3.0 inches, using 4 to 6 stitches per inch. The stitch pattern used to attach the body belt and snap hooks to the holding strap shall be a four point cross stitch, using 4 to 6 stitches per inch, with a minimum length of 2.5 inches, except that the minimum length of the four point cross stitch used to attach the snap hooks shall be 3 inches.

3.3.3 Body belt buckle and connector. The body belt buckle and connector shall be metal to metal with positive latch. The buckle and connector shall be such that the belt may be worn either as a right or left hand belt, and may be released with either hand.

3.3.3.1 Body belt buckle (Item 3). The body belt buckle shall be AM-SAFE, Inc., Part No. 501000 or equal (see 6.4). The buckle shall consist of a base and a spring loaded cover that houses the locking and adjustment mechanism. The buckle shall be designed so the release tab must be lifted at least 90 degrees perpendicular to the base prior to release of the connector. The buckle shall be fitted with a friction-type length adjustment on one side, allowing adjustment of the belt length without releasing the buckle.

3.3.3.2 Body belt connector. The body belt connector shall be AM-SAFE, Inc., Part No. 449350 or equal (see 6.4). The male end of the connector shall consist of a single slot type tongue for insertion into the locking and adjustment mechanism. In addition, the connector shall also include a slot for insertion of the nylon webbing strap that is to be looped back and sewn. There will be no adjustment mechanism on this end.

3.3.4 Snap hooks (Item 1). Snap hooks shall be of the round-nose type with a spring closed latch. Snap hook throat and throat opening shall be 5/8 inch. The nose shall overlap the keeper latch.

3.3.5 Adapters (Item 5). The inside width of the adapters shall be 1-3/4 inches.

3.3.6 Color. Color of the webbings, slide keepers, and reinforcing inserts (Items 6, 7, 12, and 13) shall be a commercially available color and shall be uniform throughout the lot. All textile materials shall be colorfast.

3.4 Performance. Belts shall be capable of successfully passing all specified tests. Total slippage of webbing through the adjusting devices shall not exceed 1 inch when tested.

3.5 Label. Each belt shall have a combination identification and instruction label. The printing shall show fastness to laundering. The label shall bear the following inscription:

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:02Z
Check the source to verify that this is the current version before use.

BELT, SAFETY, BOAT CREW
BASIC FIBER: 100% NYLON
CONTRACT NO. DLA 100-00-0-0000 (EXAMPLE)
STOCK NO. 4240-01-000-0000 (EXAMPLE)
NAME OF CONTRACTOR:

HAND LAUNDER ONLY USING MILD SOAP OR DETERGENT - RINSE COMPLETELY
DO NOT USE BLEACH OR OTHER CHLORINE PRODUCTS
HANG TO AIR DRY - DO NOT STOW WET - DO NOT IRON
AFTER IMMERSION IN SALT WATER - RINSE THOROUGHLY IN CLEAN FRESH WATER
AIR DRY BY HANGING

DO NOT REMOVE THIS LABEL

3.6 Workmanship. After completion of the final assembly, the boat crew safety belt shall be thoroughly cleaned and all thread scraps, lint and foreign matter shall be removed. Any metal or non-fibrous components shall not be misaligned, malformed, fractured, broken or distorted nor contain any sharp edge, crack, dent, nick, burr or sliver. The boat crew safety belt shall not contain any spot, stain, non-specified hole, abraded area, tear, cut, mend or needle chew. The boat crew safety belt shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability or durability. The boat crew safety belt shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established in 4.3.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2. Lot size. A lot shall consist of all belts presented at one time for the purpose of acceptance.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T21:02Z
Check the source to verify that this is the currentversion before use.

4.3 Inspection. The preproduction sample and all belts in the lot shall be subjected to inspection for the defects listed in table I, and to assure conformance to specified requirements.

TABLE I. Classification of defects

Defect	Major	Minor
Any cut, hole or tear in fabric	X	
Broken or incorrect stitching	X	
Incorrect dimension	X	
Missing, loose, broken or distorted parts	X	
Incorrect assembly	X	
Sharp edges in hardware	X	
Nylon strap ends not properly seared		X
Rust on metallic parts		X
Poor plating		X
Missing or incorrect marking		X

4.3.1 Acceptance criteria. Any belt having one or more major defects, or two or more minor defects shall be rejected. A lot containing more than 2.5 percent rejects shall be sufficient cause for rejection of the entire lot.

4.4 Acceptance test samples. Two belts shall be selected as acceptance test samples for the tests of 4.5.1, 4.5.2, 4.5.3, and 4.5.4.

4.5 Tests. The preproduction sample shall be subjected to the tests of 4.5.1, 4.5.2, 4.5.3, and 4.5.4. Both acceptance test samples from each lot shall be subjected to the tests of 4.5.1 and 4.5.2. Upon successful completion of these tests, one acceptance test sample shall be subjected to the test of 4.5.3, while hardware taken from the second acceptance test sample shall be subjected to the test of 4.5.4. Failure of any specimen to pass the tests specified shall be cause for rejection of the lot represented by the specimen.

4.5.1 Body belt test. Securely fasten the body belt around a 200-pound bag of sand and attach both snaphooks to a 1/2-inch filament nylon rope with a free length of 10 feet plus or minus 2 inches. Secure the free end of the rope line to a rigid overhead support high enough to allow a 5-foot clearance between the sand bag and the floor or ground. Allow the bag to fall freely through a distance of 5 feet plus or minues 2 inches, measured from a fixed position on the sand bag. Examine the belt to determine compliance with specified requirements.

4.5.2 Holding strap test. The holding strap shall be connected by one snap hook to a D-ring which is suitably lashed to a 200-pound bag of sand. Secure the other snap hook to a rigid overhead support high enough to allow a 5-foot clearance between the sand bag and the floor or ground. Allow the bag to fall freely through a distance of 5 feet. Repeat this test two more times. Examine the holding strap to determine compliance with the requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

4.5.3 Destruction test. One of the belts, after having passed the tests specified in 4.5.1 and 4.5.2, shall be used for the destruction test. With belt buckle fastened in service position, holding strap adjusted to a length slightly greater than the total length of the body belt, and the snap hooks fastened to each other, the body belt and holding strap of a complete belt shall be placed over 2-inch mandrels. The mandrels shall be separated at a uniform rate of not less than 1 inch nor more than 4 inches per minute. The pull on the belt shall be from the front to rear. This test shall be repeated with the holding straps adjusted to a shorter total length than the body belt. Failure occuring under a total load of 2000 pounds in any of the metal parts or stitching, or failure in the performance of the belt to meet the requirements of paragraphs 3.2.4, 3.3.2, and 3.4, respectively, shall be sufficient cause for rejection of the lot. All belts subjected to these destruction tests shall be destroyed upon completion of these tests.

4.5.4 Corrosion resistance test. Exterior metal parts of the belt assembly shall be subjected to a salt spray (fog) test in accordance with ASTM B 117, for a period of 100 hours, consisting of four periods of 24 hours exposure with one hour drying time between each period and one hour drying time upon completion of the fourth period. Immediately after completion of this exposure and drying, the parts shall be examined for compliance with the requirements of 3.2.4.

5. PACKAGING

5.1 Unit packing. Belts shall be individually boxed conforming to standard commercial practice.

5.2 Packing. Unit packs shall be packed and prepared for shipment to permit acceptance by common carrier and to afford maximum protection from normal hazards of transportation.

5.3 Marking. The interior and exterior containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year) and the contract or order number. The exterior containers shall also be marked to conform with Uniform Freight Classification Rules and National Motor Freight Classification Rules, as applicable. Each unit pack and shipping container shall be marked with the item nomenclature, quantity, contract or order numbers, manufacturer's name, and national stock number.

6. NOTES

6.1 Intended use. Belts covered by this specification are intended for personnel engaged in hazardous boat operations.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification, and the national stock number.

6.3 Material. In their quotations manufacturers shall indicate the type of material to be used for the metal parts (see 3.2.4).

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:02Z
Check the source to verify that this is the current version before use.

6.4 Specal instructions. Prior to use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval. The body belt buckle and connector specified in 3.3.3.1 and 3.3.3.2 may be obtained from AM-SAFE, Inc., 240 N. 48th Avenue, Phoenix, AZ 85043. Specify Part No. 501000 for the body belt buckle and Part No. 449350 for the associated connection.

6.5 Subject term (key word) listing.

Belt buckle
Belt connector
Boat crew belt
Safety belt

Custodian:

Navy - CG

Preparing activity:

Navy - CG

Project No. 4240-N942

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:02Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commandant (G-OSR-2)
U.S. Coast Guard
Washington, DC 20593

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-51492A(CG)	BELT, SAFETY, BOAT CREW

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITIONS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-02-22T21:02Z
Check the source to verify that this is the current version before use.